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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

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TITLE OF THE PROJECT	Smart Book Recommendation System based on User Preferences	
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Abstract:

With the rapid growth of digital libraries, e-books, and online reading platforms, users have access to thousands of books across different genres, authors, languages, and categories. Although this wide availability provides users with more choices, it also makes it difficult to identify books that match their personal interests. Traditional methods of searching for books mainly depend on keywords, categories, or general popularity and may not provide recommendations that are personalized according to individual users. Therefore, there is a need for an intelligent and personalized book recommendation system that can understand user preferences and suggest relevant books.

The proposed Smart Book Recommendation System Based on User Preferences is designed to provide personalized book recommendations by analyzing the interests and reading behavior of users. The system allows users to provide information such as their preferred genres, favorite authors, language preferences, ratings, and previously read books. Based on this information, the system analyzes the user's interests and identifies books that are likely to match their preferences. The system can also consider user ratings and feedback to improve the relevance of future recommendations.

The proposed system uses recommendation techniques such as content-based filtering and collaborative filtering. Content-based filtering recommends books that have characteristics similar to books previously preferred by the user, such as genre, author, subject, or other book attributes. Collaborative filtering identifies users with similar reading preferences and recommends books that were highly rated or preferred by similar users. By combining these approaches, the system can generate more personalized and meaningful recommendations compared with simple popularity-based or category-based suggestions.

The system can maintain user profiles and continuously update recommendations based on their interactions, ratings, searches, and reading history. Machine learning techniques can be incorporated to identify patterns in user preferences and improve recommendation accuracy over time. The recommended books can be displayed along with important details such as title, author, genre, rating, and description, allowing users to easily select books that match their interests.

The main objective of the proposed system is to reduce the time and effort required for users to search for suitable books while improving their overall reading experience. The system can help users discover new authors and books that they may not find through traditional searches. It can also provide a simple, efficient, and user-friendly platform for personalized book discovery. In the future, the system can be extended with advanced machine learning and natural language processing techniques to analyze book descriptions, reviews, and user feedback, thereby providing more accurate and dynamic recommendations.

Furthermore, the system follows a **Service-Oriented Architecture (SOA)** approach, where major functionalities are divided into independent and reusable services such as User Service, Book Service, Recommendation Service, and Rating & Review Service. An API Gateway acts as a central communication point between the user interface and these services, providing secure request routing and service access. Each service manages its own data and communicates through REST APIs, making the system scalable, modular, and easier to maintain. The architecture also allows additional services such as notification, analytics, and cloud storage to be integrated in the future without affecting the existing components. This SOA-based design enables the recommendation system to efficiently

handle increasing users and book data while supporting continuous improvement of personalized recommendations.

Project Architecture/Project Overview:

